

Homework — 2.1.4 Thinking Logically

Name: _____ Date: _____ Mark: _____ / 20

OCR H446 — set after the 2.1.4 lesson. Code uses OCR Exam Reference Language.

Part A — This week's topic (~14 marks)

1. An online exam system awards a **Distinction** for a score of 70 or above; otherwise a **Pass** for a score of 40 or above; otherwise a **Fail**. Complete the three missing parts of the program, labelled (a) to (c). [3] (AO2)

```
score = input("Enter the score")
if (a) _____ then
    print("Distinction")
elseif (b) _____ then
    print("Pass")
(c) _____
    print("Fail")
endif
```

2. Complete the table by writing the exact Boolean condition tested at each decision point, using the variable name given. [2] (AO2)

Requirement	Boolean condition
A theme-park ride admits a rider who is at least 1.4 m tall (variable <code>height</code>)	
A player earns a bonus when their score is more than 500 (variable <code>score</code>)	
A withdrawal is allowed when the amount is no more than the balance (variables <code>amount</code> , <code>balance</code>)	
A heater switches on when the temperature falls below 18 (variable <code>temp</code>)	

3. For each statement, write **True** or **False** in the middle column. If a statement is false, write a corrected version in the final column. [4] (AO2)

Statement	True / False	Correction if false
A shop gives a discount when a customer has spent £100 or more. The condition <code>spent > 100</code> awards the discount to a customer who has spent exactly £100.		
The condition <code>amount <= balance</code> OR <code>amount <= dailyLimit</code> allows a cash withdrawal only when both limits are satisfied.		
In an <code>if ... elseif ...</code> chain, the <code>elseif</code> condition is only tested when the first condition is false.		
A water-tank controller turns its pump ON when <code>level < 20</code> and OFF when <code>level > 80</code> . At a level of exactly 50% it leaves the pump unchanged.		

4. The water-tank controller is tested with five stored readings. Complete the trace table for the program below, showing one row per loop iteration. [3] (AO2)

```

    pump = "OFF"
    levels = [15, 50, 85, 70, 10]
    for i = 0 to 4
        if levels[i] < 20 then
            pump = "ON"
        elseif levels[i] > 80 then
            pump = "OFF"
        endif
        print(pump)
    next i

```

i	levels[i]	levels[i] < 20 ?	levels[i] > 80 ?	pump	Output printed

5. A bank allows a cash-machine withdrawal only if the amount requested is **no more than** the account balance **and** no more than the daily limit. Write a suitable Boolean condition using a logical operator, and state what happens when the condition is **false**. [2] (AO2)

Part B — Retrieval: keep it fresh (~6 marks)

6. (2.1.3) In a payroll program the order of steps is: read hours → calculate gross pay → deduct tax → output payslip. Explain why "deduct tax" must come **after** "calculate gross pay". [2] (AO2)

7. (1.3.2) State what is meant by a **primary key** and a **foreign key** in a relational database. [2] (AO1)

8. (1.4.1) Convert the denary number **45** into **8-bit binary**. [2] (AO2)
